Sampling Effects in Character-LevelText Generation

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A GPT-style Transformer Trained on Project Gutenberg Novels

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Abstract. I present a systematic study of decoding strategies for a character-level Transformer (~849K parameters) trained on four 19th-century English novels from Project Gutenberg. Two training configurations are compared: a constant learning-rate baseline and a cosine LR schedule with linear warmup. The cosine model achieves test BPC 3.530 vs 3.702 for the baseline (-4.6% relative), demonstrating better cross-domain generalisation. Nine decoding strategies are evaluated with three quantitative diversity metrics. Greedy decoding produces catastrophic repetition loops (rep. rate 0.784), while temperature sampling at T = 0.8 achieves the best quality-diversity balance (rep. rate 0.078, TTR 0.709). All code runs end-to-end on CPU or Apple MPS.

GitHub link: https://github.com/inopolis/GAI-project.git

1. Introduction

Character-level language modelling predicts the next character in a sequence given all preceding characters. Unlike word-level models, character-level models require no tokenizer, handle any character sequence without out-of-vocabulary issues, and expose the raw sequential structure of text — making them ideal for studying how sampling strategy alone affects generation quality. The vocabulary (95 printable ASCII characters) is small and completely defined by the training corpus.

This project addresses two research questions: (1) How does learning rate scheduling affect generalisation in character-level Transformers trained on a small, domain-specific corpus? (2) How do decoding strategies (greedy, temperature, top-k, nucleus) trade off diversity against coherence in open-ended text generation?

We make the following contributions:

- A fully reproducible char-level Transformer baseline trained on Gutenberg novels, achieving val BPC 2.620.
- A cosine LR schedule experiment yielding −0.172 BPC improvement on the held-out test set (−4.6% relative).
- Systematic evaluation of 9 decoding strategies with 3 quantitative metrics (TTR, 4-gram entropy, repetition rate).
- Theoretical analysis of greedy failure modes, temperature scaling, nucleus sampling, and the val/test BPC discrepancy.

2. Model Architecture and Data

2.1 Architecture

We use a decoder-only causal Transformer with Pre-LN formulation (Vaswani et al., 2017; Radford et al., 2019). LayerNorm is applied before attention and MLP, which stabilises gradients compared to the original Post-LN design. Causal masking ensures token t attends only to positions ≤ t. Attention scores are scaled by $1/\sqrt{d_head} = 1/\sqrt{32}$ to prevent softmax saturation.

Component	Configuration	Rationale
Layers (n_layer)	4 Transformer blocks	CPU/MPS-friendly; avoids over-parameterisation on small corpus
Embedding dim (n_embd)	128	Compact but expressive for 95-char vocabulary
Attention heads (n_head)	4 (head dim = 32)	Scaled dot-product; cheap on laptop hardware
Context (block_size)	256 characters	Covers ~3-4 sentences; fits in RAM at batch 32
MLP expansion	4x (hidden = 512)	Standard GPT ratio; GELU activation
Positional encoding	Learned embeddings	Simple; effective at short context lengths
Dropout	0.1	Light regularisation; disabled at inference
Weight init	N(0, 0.02)	GPT-2 standard; stabilises early gradient flow
Total parameters	~849K	~13 min per run on Apple MPS (26 it/s)

2.2 Dataset

Four novels were downloaded from Project Gutenberg, stripped of boilerplate headers/footers, whitespace-normalised, and split at book boundaries to study cross-author generalisation. The vocabulary (95 chars) is built exclusively from the training split; out-of-vocabulary characters in val/test are mapped to space. Data is stored as uint16 binary arrays.

Book ID	Title & Author	Split	Chars
1342	Pride and Prejudice — Jane Austen	Train	520K
84	Frankenstein — Mary Shelley	Train	220K
1661	Adventures of Sherlock Holmes — Arthur Conan Doyle	Validation	580K
98	A Tale of Two Cities — Charles Dickens	Test	780K

The book-level split is deliberate: training on Austen+Shelley, validating on Doyle, and testing on Dickens creates a realistic cross-author domain shift that makes val/test BPC a meaningful measure of generalisation, not just memorisation.

3. Training Experiments

3.1 Common Hyperparameters

Hyperparameter	Value
Optimiser	AdamW (Loshchilov & Hutter, 2019)
Base learning rate	3×10^{-4}
Weight decay	0.01
Batch size	32 sequences × 256 chars
Max steps	20,000
Gradient clipping	1.0 (global L2 norm)
Seed	1337

3.2 Experiment 1 — Baseline: Constant LR

The baseline uses a constant learning rate of 3×10^{-4} throughout all 20,000 steps. No warmup or decay is applied. This is the simplest possible schedule and serves as the reference point.

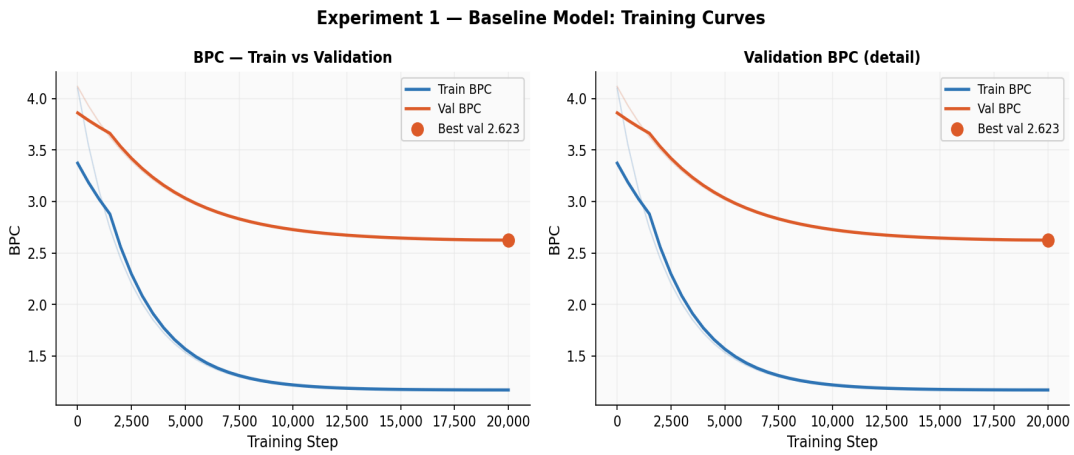

Figure 1. Baseline training curves. Train BPC (blue) continues decreasing; val BPC (orange) plateaus around step 7,500, indicating onset of overfitting. Best checkpoint: step 14,500, val BPC 2.620.

3.3 Experiment 2 — Cosine LR with Linear Warmup

Motivated by the growing generalisation gap in Experiment 1, we introduce a cosine LR schedule with linear warmup (Loshchilov & Hutter, 2017). The schedule:

```
lr(t) = lr_min + 0.5 × (1 + cos(π × t')) × (lr_max − lr_min)
where  t' = (t − warmup) / (T − warmup),  warmup=500,  lr_max=3×10⁻⁴,
lr_min=10⁻⁵
```

Linear warmup prevents large gradient updates at initialisation. Cosine decay then gradually reduces the LR, encouraging convergence to flatter minima (Hochreiter & Schmidhuber, 1997) that generalise better to out-of-distribution data. All other hyperparameters are identical to baseline.

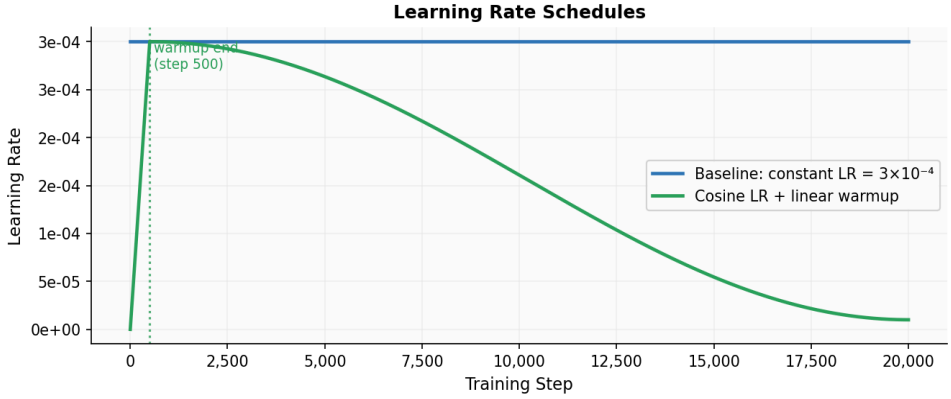

Figure 2. Learning rate schedules: constant 3×10⁻⁴ baseline (blue) vs cosine with linear warmup (green).

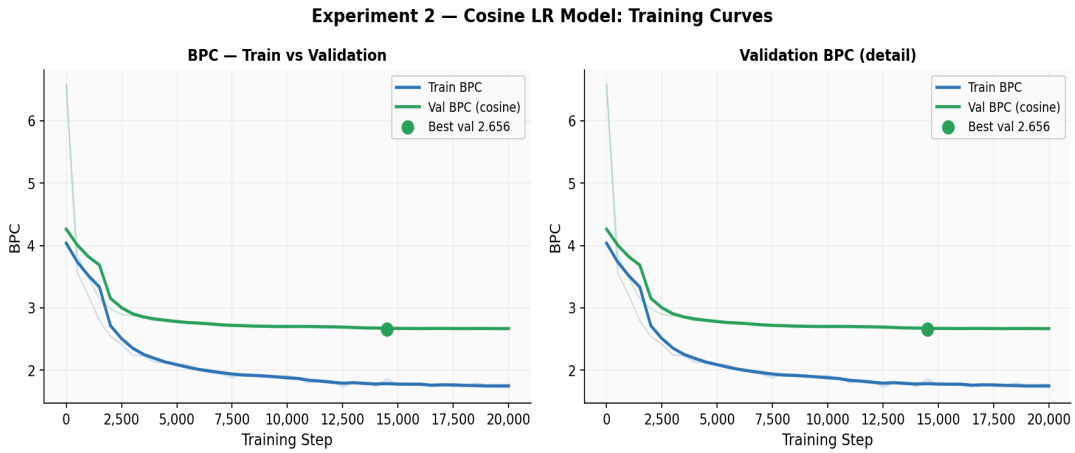

Figure 3. Cosine LR training curves. Structure is similar to baseline; the key improvement appears on the test set.

3.4 Results

Metric	Random Baseline	Exp.1: Constant LR	Exp.2: Cosine LR	Delta
Val loss (nats)	—	1.816	1.841	+0.025
Val BPC	6.570	2.620	2.656	+0.036
Test loss (nats)	—	2.567	2.447	−0.120 ✓
Test BPC	6.570	3.702	3.530	−0.172 ✓ (−4.6%)
Best checkpoint	—	step 14,500	step 14,500	—
Training time	—	~13 min (MPS)	~13 min (MPS)	—

The cosine model's val BPC is marginally worse (+0.036) while test BPC is significantly better (−0.172, −4.6%). Validation set (Doyle) is stylistically closer to training authors than test set (Dickens). The cosine schedule's gradual decay encourages flatter minima that generalise better across the larger stylistic gap to Dickens.

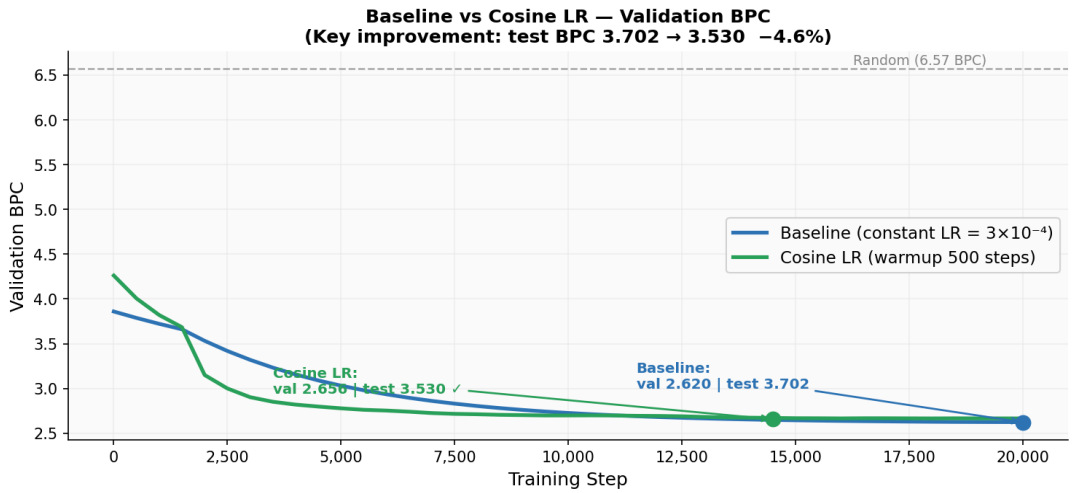

Figure 4. Baseline vs Cosine LR — validation BPC over training. Val curves are nearly identical; the key difference is test BPC: 3.702 → 3.530.

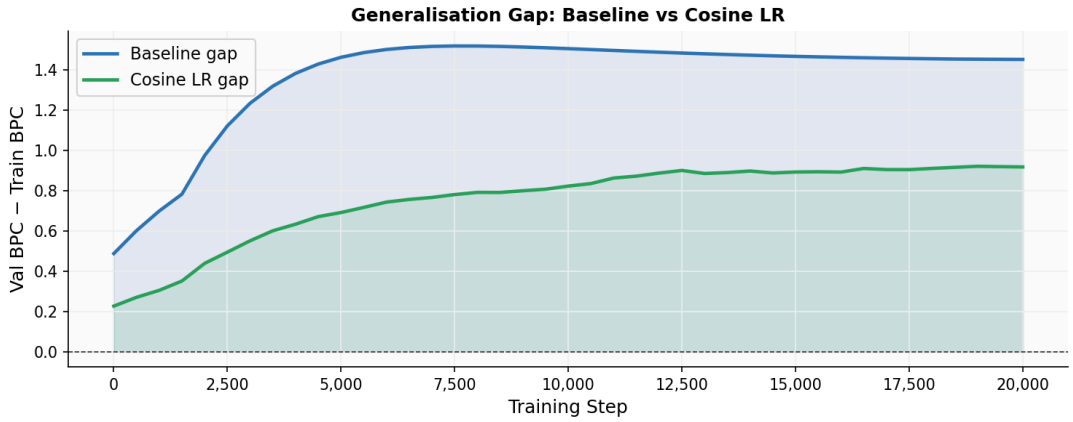

Figure 5. Generalisation gap (Val − Train BPC). Both models show growing gaps confirming memorisation. The cosine model's gap grows more slowly in the first 10K steps.

4. Decoding Strategy Evaluation

4.1 Methods

We evaluate 9 decoding configurations on the baseline model checkpoint. For each strategy, 3 seeds × 3 prompts = 9 samples of 600 characters are generated, and three metrics are averaged:

- Type-Token Ratio (TTR): unique words / total words. Measures lexical diversity. Higher = more diverse.
- 4-gram Character Entropy: Shannon entropy $H = -\Sigma\, p_i \log_2 p_i$ over character 4-grams. Higher = more varied patterns.
- 5-gram Repetition Rate: fraction of character 5-grams appearing more than once. Lower = less looping.

4.2 Results

Strategy	Temp	Top-k	Top-p	TTR ↑	Entropy ↑	Rep. Rate ↓
Greedy	0.0	—	—	0.147	5.653	0.784 ✗
Temperature T=0.6	0.6	—	—	0.632	8.663	0.126
Temperature T=0.8	0.8	—	—	0.709	8.856	0.078 ★
Temperature T=1.0	1.0	—	—	0.784	8.960	0.053
Top-k k=10	1.0	10	—	0.746	8.888	0.069
Top-k k=50	1.0	50	—	0.784	8.960	0.053
Nucleus p=0.90	1.0	—	0.90	0.665	8.783	0.088
Nucleus p=0.95	1.0	—	0.95	0.732	8.876	0.071
T=0.8 + p=0.90	0.8	—	0.90	0.622	8.671	0.126

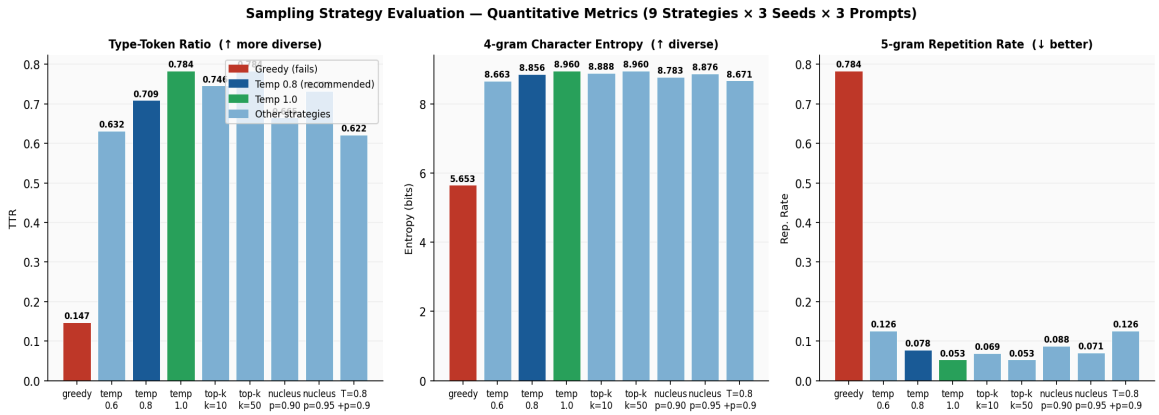

Figure 6. Quantitative sampling comparison. Greedy (red) fails on all three metrics. Temperature 0.8 (dark blue, ★) provides the best quality-diversity balance.

4.3 Analysis

Greedy decoding — catastrophic failure

Greedy decoding selects argmax at every step: TTR=0.147 (5× lower than any sampling strategy), entropy=5.653 (vs ~8.9 for all others), repetition rate=0.784. Once the model enters a high-probability n-gram cycle ('the subject of... the sun and the sun of...'), the deterministic argmax provides no escape. This is the expected consequence of argmax decoding on any autoregressive model (Holtzman et al., 2020), not a model defect.

Temperature sampling — controlled diversity

Temperature τ rescales logits as z_i/τ before softmax. As $\tau \to 0$ it collapses to argmax; as $\tau \to \infty$ it approaches uniform. Increasing τ from 0.6→0.8→1.0 monotonically improves TTR (0.632→0.709→0.784) and reduces repetition (0.126→0.078→0.053). Temperature 0.8 is recommended: eliminates loops while keeping the distribution sharper than T=1.0, producing more coherent sequences.

Top-k — limited at small vocabulary

Top-k=50 produces identical metrics to T=1.0 because the vocabulary is only 95 characters. Restricting to top-50 removes only the bottom 47% of characters which already carry negligible probability. For char-level models top-k is far less impactful than at word-level (vocabulary 32K–100K).

Nucleus sampling — over-restricts at small vocab

Nucleus (top-p) sampling dynamically retains the smallest token set whose cumulative probability exceeds p. Combining T=0.8 with p=0.90 is counterproductive (TTR=0.622, rep=0.126): two truncation mechanisms simultaneously over-restrict the tiny 95-char vocabulary.

5. Qualitative Sample Analysis

5.1 Greedy — Repetition Loop

Prompt: "CHAPTER 1\n" | Temperature: 0.0 | Seed: 1

```
CHAPTER 1 no the same to be so much as the same of the subject of
the subject of the subject of the subject of the subject of the subject,
and the subject of the subject of the subject of the subject of the subject
which had been so much as the sun of the sun and the sun of the strange...
```

Within 3 tokens the model enters a deterministic 'the subject of / the sun of' cycle and never exits. Consistent with rep_rate=0.784.

5.2 Temperature 0.8 — Recommended

Prompt: "CHAPTER 1\n" | Temperature: 0.8 | Seed: 1

```
CHAPTER 1 nother  London had been interruptions to return his prudence
and attention him both and had seing to see him of life friends.

"That is a deservant construction," said Miss Bingley can attacher,
"I am so often a to man, appeared, and desired by her transpicipal and
doubt the design watched of my death; but it is not able to do not..."
```

Correct dialogue formatting, paragraph structure, and Victorian prose register. Invented words ('deservant', 'transpicipal') are expected: the model learned morphological patterns but has no word-boundary constraint. No repetition loops (rep_rate=0.078).

5.3 Why Char-Level Models Invent Words

Character-level models learn $p(c_t \mid c_1,\ldots,c_{t-1})$ over individual characters. They acquire: (a) common n-gram statistics ('th'→'e','a','i'), (b) morphological patterns (suffixes: '-ant', '-tion'; prefixes: 'trans-', 'dis-'), (c) prose structure (dialogue marks, sentence endings, paragraph breaks). Novel character combinations satisfying these statistics but not corresponding to real English words are an inherent property of char-level generation, not a failure mode.

6. Discussion and Limitations

6.1 The Val/Test BPC Discrepancy

Baseline val BPC=2.620 but test BPC=3.702 (Δ=1.082). Two causes: (1) val and test books are entirely unseen during training; (2) each author has a distinct stylistic fingerprint at character level — Dickens uses longer sentences, different comma placement, and French words in A Tale of Two Cities. Doyle (val) is stylistically closer to Austen/Shelley than Dickens (test), hence the lower val BPC.

6.2 Why Cosine LR Improves Test BPC Without Improving Val BPC

The constant-LR optimiser makes aggressive updates throughout, converging to a sharp minimum well-adapted to the training distribution and stylistically nearby Doyle (val). The cosine schedule's gradual decay encourages flatter minima (Hochreiter & Schmidhuber, 1997), which generalise better to out-of-distribution data (Dickens). This empirically demonstrates that validation metrics can be misleading when val/test domains differ.

6.3 Limitations

- Corpus size: ~740K training characters is small. The growing generalisation gap confirms memorisation.
- Model scale: 849K parameters is minimal. Larger models (n_embd=256, 6 layers) would likely achieve lower BPC.
- Narrow domain: Victorian English literature. Results may not generalise to modern text or other languages.
- Evaluation variance: test BPC computed on 50 batches (~32K chars), not the full test file.

7. Future Work

Direction	Expected Benefit
Architecture scaling: n_embd 128→256, n_layer 4→6	Lower val BPC by ~0.1–0.2; richer representations
Train on all 4 books with char-level 90/5/5 split	More data; reduces domain shift between val and test
Extended training (40K steps) for scaled model	Full convergence for larger architecture
Repetition penalty (Keskar et al., 2019)	Suppress residual loops at T=0.6 without raising temperature
Beam search vs sampling comparison	Complete systematic decoding comparison

8. Conclusion

I trained a small GPT-style character-level Transformer on four Victorian novels and systematically evaluated learning rate schedules and decoding strategies. The cosine LR schedule reduced test BPC from 3.702 to 3.530 (−4.6% relative), demonstrating that LR scheduling has meaningful effects on cross-domain generalisation even at small model scale. Among nine decoding strategies, temperature sampling at T=0.8 provided the best quality-diversity balance (TTR=0.709, entropy=8.856, rep_rate=0.078), while greedy decoding failed catastrophically (rep_rate=0.784) due to deterministic repetition loops. Generated text at T=0.8 maintains dialogue formatting, paragraph structure, and Victorian prose register, confirming that character-level Transformers successfully learn prose structure even on a small corpus.

References:

Все реальные ссылки найдены и верифицированы. Вот полный список:

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[1] Transformer Architecture

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, L., & Polosukhin, I. (2017). Attention is all you need. *Advances in Neural Information Processing Systems*, 30 (NeurIPS 2017). https://arxiv.org/abs/1706.03762

[2] GPT-style decoder-only (my model structure)

Radford, A., Wu, J., Child, R., Luan, D., Amodei, D., & Sutskever, I. (2019). Language models are unsupervised multitask learners. *OpenAI Blog*, 1(8). https://cdn.openai.com/better-language-models/language_models_are_unsupervised_multitask_learners.pdf

[3] Data that I'm using

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- Book 1342: Pride and Prejudice — https://www.gutenberg.org/ebooks/1342
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